

Membrane Contact Probability-based Computational Studies: Sequence, Structure, and Dynamics of Membrane Proteins

Membrane proteins are essential to cellular life, yet accurate modeling often remains challenging due to their intricate interactions with the lipid bilayer. This talk highlights our recent methodological advances in membrane protein computation, centered on a metric we proposed—the membrane contact probability (MCP)—which quantifies the likelihood that an amino acid residue contacts lipid acyl chains. Building upon MCP, we have developed a series of computational tools: ProtRAP, which predicts residue accessibility of membrane proteins and enables whole-genome screening; RAPFold, which improves the structure prediction accuracy of certain membrane proteins; MemConverter, which achieves directed conversion of soluble proteins into membrane proteins through iterative sequence design; and MCP-ANM, an anisotropic network model that incorporates membrane effects and captures the constraints imposed by lipids on protein dynamics. Collectively, this series of work establishes a computational framework spanning membrane protein sequence, structure, and dynamics, which may offer new strategies for drug development and membrane protein design.